

3 In plants and humans, the phenotype of an organism is determined by the genotype and the environment.

(a) Plants from the genus *Primula* have different petal colours. The presence of the pigment malvidin results in blue petals.

The metabolic pathway for malvidin synthesis is controlled by gene *T/t*. The presence of the dominant allele **T** results in blue petals.

Another gene, gene *D/d*, at a different locus, also influences the malvidin synthesis pathway. When the dominant allele **D** is present, its gene product suppresses the malvidin synthesis pathway. This is summarised in Fig. 3.1.

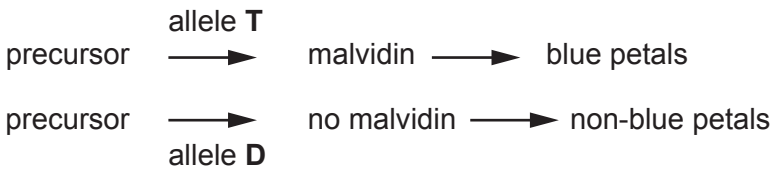


Fig. 3.1

A genetic cross was carried out between two plants heterozygous at both gene loci. The resulting offspring genotypes are shown in a Punnett square in Fig. 3.2.

	TD	Td	tD	td
TD	TTDD	TTDd	TtDD	TtDd
Td	TTDd	TTdd	TtDd	Ttdd
tD	TtDD	TtDd	ttDD	ttDd
td	TtDd	Ttdd	ttDd	ttdd

Fig. 3.2

- (i) State which of the genotypes shown in Fig. 3.2 have blue petals.

 [1]
- (ii) State the ratio of non-blue to blue petals for the cross shown in Fig. 3.2.
 [1]
- (iii) Name the type of gene interaction that has caused the offspring ratio you have stated in (a)(ii).
 [1]

- (iv) Gene ***T/t*** and gene ***D/d*** code for proteins that are involved in the control of the production of malvidin.

Discuss the possible roles of the proteins coded for by gene ***T/t*** and gene ***D/d*** in the control of the production of malvidin.

[5]

- (b)** Some humans have the inherited condition haemophilia.

Explain the relationship between the **F8** gene, factor VIII and the condition haemophilia.

..... [4]